

Rojashri Goluguri

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Education

Yeshwantrao Chavan College of Engineering

Bachelor of Technology in Electronics and Telecommunication, Minor in Data science

Nagpur, Maharashtra

August 2023 - June 2027

Motoshri kashidevi vdy, Musewadi

11th - 12th

Musewadi, Nagpur, Maharashtra

August 2021 - April 2023

Experience

Intern

Summer School at VNIT

June 2025 - July 2025

Nagpur, Maharashtra

- Performed hands-on experiments in circuit design, testing, and troubleshooting of electronic components and systems.
- Collaborated with faculty and peers on technical projects, enhancing problem-solving abilities and gaining exposure to real-world engineering practices.
- Developed an IoT-based Heart Rate Monitoring System using ESP32 with real-time data logging and visualization.
- Assisted in circuit analysis and simulation using tools like GNU Radio and documented results using Overleaf for performance evaluation and reporting.

Projects

Active LiDAR-Prioritized Surveillance System | LiDAR, ESP32, Python, IoT Alerts

January 2026 – Present

- Implementing a priority-based detection algorithm to identify the closest/high-risk target and trigger surveillance response.
- Integrating LiDAR sensor with the controller to perform real-time scanning and obstacle/object detection.
- Developing logic for automatic alert generation when an object enters the restricted zone.
- Improving monitoring efficiency by reducing false detections and focusing on high-priority targets.
- Testing and validating system performance under different distance ranges and environmental conditions.

IOT based Smart Shopping cart using RFID | ESP32, RFID, Embedded C, IoT/WiFi

October 2025

- Built an RFID-based shopping cart to automate item scanning and billing without manual barcode entry.
- Designed embedded logic to update cart items dynamically and compute total price in real time.
- Integrated LCD interface for displaying scanned item details and final billing amount.
- Implemented IoT-based synchronization to send billing data to a backend/cloud platform.

5G based railway track monitoring system using ESP32 | HTTP, Data Visualization

June 2025 - July 2025

- Developed a heart rate monitoring system using optical pulse sensing to measure blood flow variations and calculate BPM.
- Programmed ESP32 to process sensor signals and transmit readings via HTTP, logging live data into Google Spreadsheet.
- Built a web/dashboard application to visualize real-time BPM graphs and trigger warning notifications when BPM exceeds safe thresholds.

Room Light Controller using 8051 Microcontroller | Embedded C, Keil uVision

April 2025

- Developed an automatic room light controller using 8051 by interfacing IR sensor (motion detection) and LDR (light intensity sensing).
- Implemented Embedded C logic to turn ON lights only when motion is detected and ambient light is low.
- Added timer-based auto OFF feature to switch OFF lights after no motion is detected for a fixed duration.

Technical Skills

Languages: Java, Python, SQL, HTML

Developer Tools: Visual Studio, IntelliJ, Kaggle, matlab, Arduino IDE

Libraries: pandas, NumPy, Matplotlib